

Adaptive Demand Allocation Architecture (ADA)

System Overview — Revised

Nebico Oil Storage — Palm Oil Retention & Melting System

Based on the *VCH Sizing Framework, Second Edition*
A Unified Design Methodology for Submerged Hydronic Coil Systems

Author Sambridda Ranabhat
Organisation Mechatronics Engineering Pvt. Ltd. (MEPL)
Location Kathmandu, Nepal
Date July 16, 2026
Status Design Proposal — For Review

This document presents the revised system overview of the Adaptive Demand Allocation Architecture (ADA), covering thermofluid analysis, coil geometry, hydraulic sizing, NTU performance, the need-array priority model, and the valve/VFD mapping. Full PLC ladder logic and control implementation are documented separately in the companion document *ADA: PLC Implementation — Full Replacement*.

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Executive Summary

This document presents the system overview for the most recent revision of the Adaptive Demand Allocation Architecture (ADA): the coil bore across every tank is reduced from 2 inch to 1.5 inch, the branch valve count drops from 8 to 6, and the need array is expanded so that all **three** priorities (P1, P2, P3) can be genuinely concurrent. Two of the three branches now serve the T_{10} melting duty continuously and permanently; the third branch is shared between T_5 (P1) and the PO tank (P2), with the trunk capped at 1.2 m s^{-1} so that three-way concurrency throttles every active branch below its individual design flow rather than starving any one process.

This document covers the thermofluid analysis, architecture, and hardware mapping. Full PLC implementation — ladder logic, register map, AI/DI/DO assignments, safety gating, and all control blocks — is in the companion document: **Adaptive Demand Allocation Architecture (ADA): PLC Implementation — Full Replacement**.

Headline Results

Metric	Result
Coil area lost to the bore reduction (all tanks)	−20.0%
T_5 (P1) daily duty period	4.47 hr (18.6%)
PO tank (P2) daily duty period	0.71 hr (2.96%)
Shared-branch active fraction of the day	21.0%
Palm oil melted per day (P3)	≈4453 kg/day
Combined system throughput (P3 + PO tank)	≈4953 kg/day
Fulfilment vs. 2 t/day mandate (melt / combined)	≈222.7% / ≈247.7%
Total daily thermal energy demand	≈990.7 MJ
Average continuous power demand	≈11.47 kW
Peak instantaneous demand (worst-case)	≈36.61 kW
Auxiliary array rated capacity	36.0 kW

Key Findings

- **The bore reduction, taken alone, costs the low-NTU duties (T_5 and T_{10} melting) close to their full 20 % area loss** — delivered heat tracks lost area almost 1:1 when $NTU \ll 1$. The PO tank, sitting at meaningfully higher NTU, is structurally buffered and loses proportionally less.
- **The three-way flow split, isolated from the bore reduction, costs remarkably little** — every duty retains 98–99 % of its own dual-branch delivery rate when throttled to three-way concurrent flow.
- **Combined throughput falls by ≈8.6 %** relative to the prior (2-inch, two-priority) revision, but still clears the original 2 t/day mandate by **≈2.5×**.
- **At the single worst-case instant where every process is simultaneously at its coldest, highest-draw condition, peak demand (≈36.61 kW) marginally**

exceeds the auxiliary array’s rated 36 kW — a genuine, if narrow, finding worth carrying into the electrical design phase (Section 9).

1 Scope and Motivation of the Minor Revision

Architectural Revision II fixed the coil at 2 inch bore across every tank and allowed the need array to serve at most two branches at once. Two further hardware changes are introduced here:

1. **Coil bore reduced to 1.5 inch** across every tank (T_5 , T_{10} , PO), coil lengths unchanged (10 m for T_5 , 30 m for T_{10} /PO).
2. **Branch valve count reduced from 8 to 6**, and the need array is expanded so that **all three priorities can draw simultaneously**, not just two — the trunk is capped at a safe 1.2 m s^{-1} rather than being sized to guarantee every branch its individual full design flow under three-way concurrency.

The second change is the more consequential one architecturally: it retires the three-way “discard the lowest priority” contention rule from the PLC arbiter in favour of genuine three-way service, at a throttled flow rate, whenever all three priorities have live demand at once. The full ladder implementation of this change is covered in the companion PLC document.

2 Revised Coil Geometry

Table 1. Coil geometry, 2-inch (prior revision) vs. 1.5-inch (this revision).

Parameter	T_5 (10 m)		T_{10}/PO (30 m)	
	2-inch	1.5-inch	2-inch	1.5-inch
d_o (mm)	60.33	48.26	60.33	48.26
d_i (mm)	54.79	42.72	54.79	42.72
d_o/d_i	1.1011	1.1297	1.1011	1.1297
$R_{\text{wall}} (\times 10^{-4} \text{ m}^2 \cdot \text{K} \cdot \text{W}^{-1})$	1.782	1.805	1.782	1.805
Coil surface area (m^2)	1.895	1.516	5.685	4.549
Area change	−20.0%		−20.0%	

3 Trunk Reconfiguration: True Three-Way Concurrent Flow

3.1 Flow Split at the Trunk's Own Velocity Ceiling

The 1.5-inch trunk (area $1.433 \times 10^{-3} \text{ m}^2$) is capped at its own safe rated velocity of 1.2 m s^{-1} . Under three-way concurrency, the trunk runs at its own ceiling and the branches share what it can deliver:

$$F_{\text{trunk}} = 1.433 \times 10^{-3} \times 1.2 = 1.7196 \times 10^{-3} \text{ m}^3 \text{ s}^{-1} \quad (1)$$

$$F_{\text{branch, triple}} = \frac{F_{\text{trunk}}}{3} = 5.732 \times 10^{-4} \text{ m}^3 \text{ s}^{-1} \Rightarrow V_{\text{branch, triple}} \approx 0.944 \text{ m s}^{-1} \quad (2)$$

Single- and dual-branch operation are unaffected by this change. A single active branch or two active branches together still draw at most $2 \times 7.283 \times 10^{-4} \text{ m}^3 \text{ s}^{-1}$, well inside the trunk's own 1.2 m s^{-1} ceiling — those branches continue to receive their full individual 1.2 m s^{-1} design rate exactly as in Revision II. The throttled regime in this document applies **only** to the moment all three branches draw simultaneously.

Note on a minor cross-reference inconsistency. The branch flow area used above ($6.069 \times 10^{-4} \text{ m}^2$) traces to the project's own hand-derived trunk/branch sizing notes; the original base VCH proposal's 1-inch supply pipe basis instead implies $\approx 5.575 \times 10^{-4} \text{ m}^2$ (26.64 mm ID). This $\approx 9\%$ area discrepancy affects only the *reported branch velocity* figure — every thermal result in this document is derived from **volumetric flow rate** at the trunk split, which is unaffected. Worth reconciling against as-built pipe specification.

3.2 Capacity Rate Under Both Regimes

$$C_{w, \text{single/dual}} = 2740.7 \text{ W K}^{-1} \text{ (unchanged, Revision II basis)} \quad (3)$$

$$\dot{m}_{w, \text{triple}} = 977.8 \times 5.732 \times 10^{-4} = 0.5605 \text{ kg s}^{-1} \Rightarrow C_{w, \text{triple}} = 2348.6 \text{ W K}^{-1} \quad (4)$$

A 14.3% drop in C_w under three-way concurrency, relative to the single/dual regime.

4 Water-Side Hydraulics at the New Bore

Water at 70 °C: $\rho = 977.8 \text{ kg m}^{-3}$, $C_p = 4190 \text{ J kg}^{-1} \text{ K}^{-1}$, $\mu = 4.04 \times 10^{-4} \text{ Pa s}$, $k = 0.663 \text{ W m}^{-1} \text{ K}^{-1}$, $\text{Pr} = 2.55$.

Table 2. Reynolds number and Gnielinski friction factor, 1.5-inch bore.

Parameter	Single/dual ($\dot{m} = 0.654 \text{ kg s}^{-1}$)	Triple ($\dot{m} = 0.5605 \text{ kg s}^{-1}$)
Re_{coil}	48,251	41,349
$f = (0.79 \ln Re - 1.64)^{-2}$	0.02113	0.02190
Nu_{str}	203.2	178.75

Table 3. Inner heat transfer coefficient by tank and flow regime, 1.5-inch bore.

Tank	Single/dual ($\text{W/m}^2\text{.K}$)	Triple ($\text{W/m}^2\text{.K}$)
$h_{i,T10/PO}$ ($D_{\text{helix}} = 2.072 \text{ m}$)	3384	2978
$h_{i,T5}$ ($D_{\text{helix}} = 1.711 \text{ m}$)	3432	3020

5 Revised Overall U -Values

$$\frac{1}{U} = \frac{1}{h_i} \cdot \frac{d_o}{d_i} + R_{f,i} \cdot \frac{d_o}{d_i} + R_{\text{wall}} + R_{f,o} + \frac{1}{h_o} \quad (5)$$

using the same fouling basis as every prior revision ($R_{f,i} = 1.0 \times 10^{-4}$, $R_{f,o} = 9.0 \times 10^{-4} \text{ m}^2\text{.K.W}^{-1}$).

Table 4. Overall U -values, 1.5-inch bore, all duties and regimes.

Duty	h_o ($\text{W/m}^2\text{.K}$)	U single/dual ($\text{W/m}^2\text{.K}$)	U triple ($\text{W/m}^2\text{.K}$)
T ₅ (liquid)	138.9	115.59	115.13
PO tank (liquid)	138.9	115.55	115.08
T ₁₀ melt (solid)	35	33.28	33.23

Correction to an earlier working figure. The PO tank's single/dual-branch U -value was previously stated as $128.7 \text{ W m}^{-2} \text{ K}^{-1}$ — an arithmetic slip. The correct value is $115.55 \text{ W m}^{-2} \text{ K}^{-1}$, essentially identical to T₅'s own value. This affects only the single/dual-branch comparison row for the PO tank; all duty-period, throughput, and energy figures were already built from the correct triple-branch value.

6 NTU, Effectiveness, and Delivered Heat Rate

$$NTU = \frac{UA}{C_w}, \quad \varepsilon = 1 - e^{-NTU}, \quad \varepsilon C_w = \text{delivered rate per unit } \Delta T \quad (6)$$

Table 5. Full comparison: 2-inch (Revision II baseline) vs. 1.5-inch, both flow regimes. All εC_w values in W/K.

Duty	εC_w , 2" baseline	εC_w , 1.5" single/dual	Δ vs. baseline	εC_w , 1.5" triple	Δ vs. 1.5" single/dual
T ₅	205.3	169.75	−17.3%	168.28	−0.9%
PO tank	570.5	478.65	−16.1%	469.6	−1.9%
T ₁₀ (solid)	181.5	147.38	−18.8%	146.42	−0.7%

Two independent effects, cleanly separable. Reading left-to-right within the 1.5-inch columns isolates the **bore reduction** (baseline → single/dual, roughly 16–19 % down — tracking the 20 % area loss closely for the low-NTU duties). Reading single/dual → triple isolates the **three-way flow-split penalty** alone: under 2 % for every duty. The coil now delivers meaningfully less than before regardless of branch count, but adding the third concurrent branch costs almost nothing further.

7 The Need-Array Expansion to True Three-Way Concurrency

7.1 Architecture

Two of the three branches are now permanently and continuously dedicated to T₁₀ palm oil melting (P3) — there is no longer a scenario in which P3 is fully idle. The third branch is shared between P1 (T₅) and P2 (PO tank). The structural consequence: **any time P1 or P2 has live demand, the trunk is necessarily supplying all three branches at once** — P1 and P2 never see the better single/dual-branch rate; they always operate under the throttled triple-branch condition.

This retires the “discard the lowest priority” three-way contention rule from the prior PLC arbiter. That rule existed specifically to resolve genuine three-way contention when only two branches were physically available. It is now obsolete: three-way contention is the trunk’s normal operating condition whenever P1 or P2 is active, resolved by *flow-sharing*, not by *discarding* a priority. The ladder implementation of the new arbiter is in the companion PLC document, Section 2.8.

7.2 Timing

Using T₅’s daily demand basis (4 t/day, unchanged) and the PO tank’s (500 kg/day, unchanged), evaluated **entirely at the triple-branch rate** since that is the only regime P1/P2 ever see:

$$t_{P1} = \frac{74.4 \times 10^6}{168.28 \times 27.5} \approx 16\,077\text{ s} \approx \mathbf{4.47\text{ hr}}\text{ (18.6\%)} \quad (7)$$

$$t_{P2} = \frac{51.7 \times 10^6}{469.6 \times 43} \approx 2561\text{ s} \approx \mathbf{0.71\text{ hr}}\text{ (2.96\%)} \quad (8)$$

$$t_{\text{shared,active}} \approx t_{P1} + t_{P2} - \frac{t_{P1} \times t_{P2}}{24} \approx 4.47 + 0.71 - 0.13 \approx \mathbf{5.05\text{ hr}}\text{ (21.0\%)} \quad (9)$$

$$t_{\text{shared,idle}} = 24 - 5.05 = \mathbf{18.95\text{ hr}}\text{ (79.0\%)} \text{ — both melting branches at full single/dual rate} \quad (10)$$

8 P3 Output Under Mixed-Regime Operation

$$Q_{\text{full}}(\text{per tank, } 36^\circ\text{C}) = 147.38 \times 34 = 5011\text{ W} \quad (11)$$

$$Q_{\text{reduced}}(\text{per tank, } 36^\circ\text{C}) = 146.42 \times 34 = 4978\text{ W} \quad (12)$$

$$P_{P3} = 0.790 \times (2 \times 5011) + 0.210 \times (2 \times 4978) \approx 10.007\text{ W} \quad (13)$$

$$E_{P3} \approx 864.6\text{ MJ/day} \Rightarrow m_{P3} = \frac{864.6 \times 10^6}{194,200} \approx \mathbf{4453\text{ kg/day}} \quad (14)$$

9 Maximum Instantaneous Wattage Analysis

9.1 Motivation

Every duty-period and throughput figure above is a *time-averaged* result. Sizing the auxiliary electric array requires knowing the **worst-case simultaneous** demand — the single instant every process could plausibly draw its individual maximum at once.

9.2 Worst-Case Coincidence

The most conservative physically-plausible instant: T₅ at 40°C (coldest, highest-draw); the PO tank at 24°C (likewise); both melting branches at the 36°C melt point. Since both P1 and P2 are active at this moment, all branches are in the throttled triple-branch regime.

$$Q_{T5,\text{max}} = 168.28 \times (70 - 40) = 5048\text{ W} \quad (15)$$

$$Q_{PO,\text{max}} = 469.6 \times (70 - 24) = 21.602\text{ W} \quad (16)$$

$$Q_{T10,\text{max}} \times 2 = (146.42 \times 34) \times 2 = 9957\text{ W} \quad (17)$$

$$P_{\text{peak}} = 5048 + 21,602 + 9957 \approx \mathbf{36,607 \text{ W (36.61 kW)}} \quad (18)$$

This exceeds the auxiliary array’s rated 36 kW capacity by $\approx 1.7\%$. This is a compounded worst-case figure — every process simultaneously at its coldest condition — and is deliberately conservative. The 300 L active thermal buffer would absorb a brief coincidence as a small temperature dip rather than an outright shortfall, but the margin is narrow enough to warrant an explicit check in the electrical design phase.

10 Full Before/After Comparison

Table 6. Architectural Revision II vs. this revision.

Metric	Revision II (2", 2-priority)	This revision (1.5", 3-priority)
T_5 duty period	3.67 hr (15.3%)	4.47 hr (18.6%)
PO tank duty period	0.585 hr (2.4%)	0.71 hr (2.96%)
P_3 melt output	4919.8 kg/day	4453 kg/day (−9.5%)
Combined throughput	5419.8 kg/day	4953 kg/day (−8.6%)
Fulfilment (melt/combined)	246% / 271%	222.7% / 247.7%
Total daily thermal energy	1081.5 MJ	990.7 MJ (−8.4%)
Average power demand	12.5 kW	11.47 kW (−8.2%)
Peak instantaneous demand	Not evaluated	36.61 kW
Branch valve count	8	6
Max concurrent priorities	2	3

11 Architecture Overview: Adaptive Demand Allocation Architecture (ADA)

As of this revision, the plant’s control logic is formalised under a single name — the **Adaptive Demand Allocation Architecture (ADA)** — resting on four pillars restated here in their current, fully updated form.

11.1 Pillar 1: The Need-Array Priority Framework

Active demand is evaluated as a parallel need array across three tiers: *unbounded, guaranteed* (T_5 , P1), *bounded, operator-supervised* (PO tank, P2), and *continuous, dedicated* (palm oil melting, P3 — now permanently allocated two branches). All three tiers can be genuinely concurrent: the trunk, capped at 1.2 ms^{-1} , serves whichever combination of branches has live demand, throttling flow rather than excluding a process (Section 7).

11.2 Pillar 2: Demand-Driven Dynamic Thermal Sizing

Unchanged in principle from Revision II: aggregate instantaneous demand $E_{\text{total}} = E_{\alpha} + E_5 + E_{C^*}$ is evaluated on a 10s refresh window and converted to a dynamic water target. This revision adds the **peak instantaneous coincidence** analysis of Section 9, which stress-tests the auxiliary array against the worst physically-plausible simultaneous demand across all three duties. The T_{water} control law and dynamic demand register are fully ladderred in the companion PLC document, Section 2.5.

11.3 Pillar 3: Load-Responsive Operational Postures

Unchanged: T_5 ’s own level (α_S) governs a four-tier hysteresis ladder (Eco / Efficiency / Standard / Full Throttle) for P3’s operating posture, with mode selection feeding E_{total} causally. Mode register ladder logic is in the companion PLC document, Section 2.5.

11.4 Pillar 4: Layered Operator-Trust Governance

Unchanged: the PO tank’s suspicion meter escalates on repeated $2 \times n_d$ overrun events rather than cumulative daily volume, with autonomous isolation reserved for the third such event within a 24-hour window. Suspicion meter ladder logic and the HMI dispense interface are in the companion PLC document, Sections 2.5 and 2.7.

Architectural impact, updated. ADA now delivers $\approx 247.7\%$ of the original 2 t/day mandate — down from Revision II’s 271 %, but achieved with 2 fewer control valves and a control architecture that no longer needs to discard any priority under contention. The isolated three-way flow-split penalty confirms that this concurrency gain was essentially free in thermal terms; the throughput reduction this revision accepts was spent almost entirely on the bore reduction.

12 PLC Valve Mapping

12.1 Valve Identity

Table 7. Valve numbering and type.

Tag range	Count	Type / assignment
SV100 – SV105	6	Solenoid (water routing): SV100= T_5 , SV101=TA, SV102=TB, SV103=TC, SV104/SV105=solar selector pair
PBV100 – PBV103	4	Pneumatic (oil dispense): one per tank, T_5 /TA/TB/TC

SV and PBV numbering ranges overlap (e.g. “100” appears in both) but refer to physically distinct valves on separate lines with separate PLC output terminals — this is a shared tank-identifier convention, not a shared device.

12.2 Solar Selector Pair (SV104/SV105)

Exactly one of SV104/SV105 is closed at any time, never both, never neither: SV104 (Y15) closed $\rightarrow$ through-solar; SV105 (Y16) closed $\rightarrow$ bypass-solar. This is the hardware implementation of the θ_{Sol} vs. T_{water} gating decision: $\theta_{\text{Sol}} > T_{\text{water}} \rightarrow$ through-solar; $\theta_{\text{Sol}} < T_{\text{water}} \rightarrow$ bypass. The automatic gating ladder is in the companion PLC document, Section 2.9.

12.3 Water Routing Table (Solenoid)

NOV (number of valves) = (explicit closed-valve count) +1, counting the SV104/SV105 pair as a single unit — confirmed against every row below with no exceptions.

Table 8. Water routing: line condition, valves closed, outputs energised.

#	Condition	NOV	Closed valves
1	Water T ₅	4	101, 102, 103, (104,105)
2	Water TA	4	100, 102, 103, (104,105)
3	Water TB	4	100, 101, 103, (104,105)
4	Water TC	4	100, 101, 102, (104,105)
5	Water T ₅ +TA	3	102, 103, (104,105)
6	Water T ₅ +TB	3	101, 103, (104,105)
7	Water T ₅ +TC	3	101, 102, (104,105)
8	Water TA+TB	3	100, 103, (104,105)
9	Water TA+TC	3	100, 102, (104,105)
10	Water TB+TC	3	100, 101, (104,105)
11	Water TA+TB+TC	2	100, (104,105)
12	Water TA+TB+T ₅	2	103, (104,105)
13	Water TA+T ₅ +TC	2	102, (104,105)
14	Water T ₅ +TB+TC	2	101, (104,105)

12.4 Oil Distribution Table (Pneumatic)

Table 9. Oil distribution: output port, valves closed, outputs energised.

#	Port	NOV	Closed valves
15	Oil x	2	PBV101, (104,105)
16	Oil y	3	PBV100, PBV103, (104,105)
17	Oil z	3	PBV100, PBV102, (104,105)

On the x/y/z port logic. The mapping above is confirmed directly and cross-checked against the project's P&ID; it is taken as accurate as given and not re-derived from first principles in this document.

13 VFD Control

Both VFDs are controlled via discrete multi-speed (preset) digital inputs rather than RS485/Modbus, given that each requires only a small, fixed number of operating states:

- **VFD-01 (trunk):** 4 states (Off / LSM / HSM / FSM), each mapped to a pre-programmed target frequency. LSM = 1 branch active, HSM = 2 branches, FSM = 3 branches (full supply mode), selected via mutually-interlocked PLC digital outputs Y6/Y2/Y17.
- **VFD-02 (oil pump):** 2 states (Off / On-at-safe-speed), a single pre-programmed frequency ceiling to keep oil-side pipe velocity within safe bounds, driven via Y3.

Per the VFD manufacturer’s terminal documentation, confirm whether RUN/FWD is a command independent of speed-select, or combined into the same terminal logic, before finalising the interlock wiring — this determines whether 2 or 3 discrete lines are required for VFD-01.

14 PLC Implementation Overview

Companion document. Full PLC ladder logic, register assignments, AI/DI/DO tables, safety gating, and all control blocks are documented separately in: **Adaptive Demand Allocation Architecture (ADA): PLC Implementation — Full Replacement** (MEPL, Sambridda Ranabhat). This section provides a brief summary of what that document contains and how it maps to the architecture described above.

The companion PLC document is structured in dependency order — foundational blocks first, blocks that consume their outputs after. Its contents map to this document as follows:

Table 10. Companion document section map.

PLC Doc. Section	Content and link to this document
§2.1 Analog Inputs (AI)	11-channel AI assignment (α , θ , flow); scaling pattern for 4–20 mA channels
§2.2 Digital Inputs (DI)	X0–X10 confirmed from schematic sheet 6/10; NC polarity convention; float switch warning logic
§2.3 Digital Outputs (DO)	Y0–Y25 confirmed; resolves the Y20/HE3 naming conflict; confirms 3×12 kW array
§2.4 Safety / Mode Control	E-STOP latch, System OK flag, Automation Enable; manual/auto mutual-exclusion gating pattern
§2.5 Calculated Registers	10s refresh oscillator; T_{water} law (Pillar 2); mode hysteresis ladder (Pillar 3); suspicion score (Pillar 4); auxiliary staging; tank scoring ($S_{\text{casual}}/S_{\text{immediate}}$); height-to-mass conversion
§2.6 T_f Register	Thermal blending interlock for P1; 32-bit fixed-point ladder; variable m_p open item
§2.7 P1 Block	T_5 heat-demand flag (SV100/Y11); T_f interlock action; batch-release $\rightarrow$ AODD pump (Y5)
§2.8 P2 Block	PO tank designation (D50/D51, M0–M3); dispense pulse logic; suspicion overrun trigger; P2 demand flag
§2.9 Need-Array Arbiter	Four-destination model; branch count $\rightarrow$ LSM/HSM/FSM (Y6/Y2/Y17); four-way contention tie-break; maps directly to Section 7 and valve table in Section 12
§2.10 Pump Control	VFD-01/02, Jacket, AODD — manual and auto paths
§2.11 Solar / Auxiliary	θ_{Sol} vs. $T_{\text{water}} \rightarrow$ SV104/SV105; HE1/HE2/HE3 staging; maps to Section 13
§2.12 Fault Detection	Physics-based temperature rate-of-change alarm; level/mass rate anomaly (activity-gated); T_{10} high-limit hard alarm
§2.13 Summary / Tag Map	Complete D, T, M register address table; consolidated alarm and interlock list; two-interface operator model (physical panel + touchscreen HMI)

Register addressing is illustrative throughout the companion document. All D/T/M register numbers used there are placeholders chosen to avoid intra-document collisions; final addresses must be coordinated as one complete map before programming. Several open items also remain recorded in the companion document: the E-STOP reset input, the m_p acquisition method, the mode power-up default, tie-break behaviour on equal scores, and manual-mode LSM/FSM access.

15 System Estimation and Conclusions

1. **The bore reduction and the concurrency expansion are thermally separable, and only one of them is expensive.** The 2 inch→1.5 inch bore change costs 16–19 % of delivered heat across every duty; adding genuine three-way concurrency on top of that costs under 2 % further. The control-complexity benefits of true three-way concurrency were, in thermal terms, essentially free.
2. **The system still clears the original mandate by $\approx 2.5\times$ (247.7 % combined),** comfortably absorbing the ≈ 8.6 % throughput reduction this revision accepts.
3. **The peak instantaneous demand check is new to this revision and worth carrying forward.** At 36.61 kW against a 36 kW rated auxiliary array, the margin is narrow enough to inform the electrical design phase explicitly.
4. **The three-way need-array arbiter is architecturally simpler than Revision II's, not more complex** — flow-sharing under a velocity cap replaces the discard-lowest-priority contention rule entirely.
5. **PO tank governance and mode-selection logic (Pillars 3 and 4) are unaffected by this revision** and carry forward from Architectural Revision II. Full ladder implementations are in the companion PLC document.

Prepared by **Sambridda Ranabhat** · Mechatronics Engineering Pvt. Ltd. (MEPL),
Kathmandu, Nepal

Based on the *VCH Sizing Framework, Second Edition*
DOI: 10.5281/zenodo.21009246